

Encoder Principle

1. Introduction to Encoder

An encoder is a sensing device that provides feedback. Encoders convert motion to an electrical signal for measuring speed or motion information.

Based on the output data type, encoders can be divided into incremental encoders and absolute encoders. Based on the detection principles, there are optical, magnetic, inductive, and capacitive types.

2. Encoder Classification

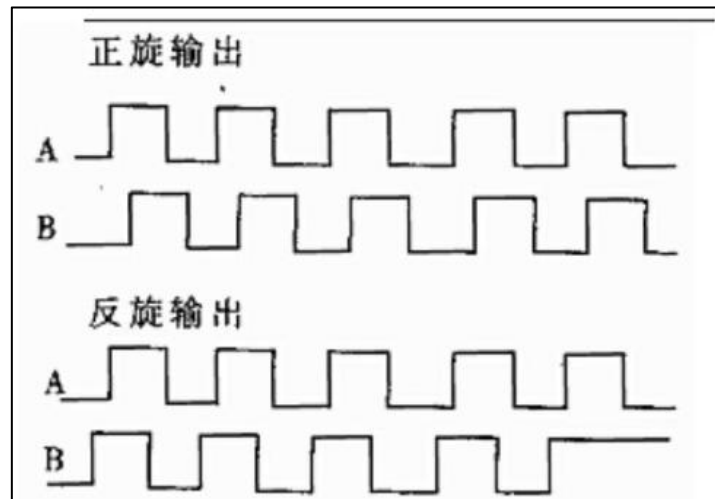
2.1 Incremental encoder

◆ Basic principle

The incremental encoder is a device that converts the displacement into a periodic electrical signal and then converts this electrical signal into counting pulses, which are then transformed into counting pulses to represent the magnitude of the displacement.

◆ Basic concept

There are five wires in total, including A, B and C three data wires, as well as VCC and GND. Both cable A and B are data output cables, outputting pulse signals for data processing. The pulses output by A and B have a phase difference of 90 degrees. The Z wire is used for the zero point signal, e.i, when encoder is rotating to the zero position, it will emit a pulse for indicating this position predetermined by manufacturer, and this position is also known as the mechanical zero position.



2.2 Absolute encoder

◆ Basic principle

Different from incremental encoder that rely on counting pulses and determining the direction of pulses, absolute encoder is a sensing device that directly output digits. More precisely, they provide a complete numerical output corresponding to each angular position. Each position is unique, and during the rotation of the encoder, the position can be directly read.

◆ Single-turn Absolute Encoder

Single-turn encoders measure displacement across 360° from a defined starting position with the output repeated for every revolution of the encoder's shaft.

◆ Multiple-turn Absolute Encoder

Multi-turn encoders measure the degree of rotation within 360° and track the number of total revolutions of the encoder's shaft using a unique word for each position and number of revolutions.

2.3 Hall Encoder



Hall encoder consists of a Hall effect disk and Hall effect sensors. The Hall disk is a circular plate with magnetic poles arranged evenly along a certain diameter. The Hall effect disk is mounted coaxially with an electric motor. When the motor rotates, the Hall effect sensors detect and output a series of pulse signals. To determine the direction of rotation, the encoder typically outputs two sets of square wave signals with a certain phase difference between them.

3. Encoder Speed Measurement

3.1 Basic Principle

When a DC motor with Hall sensor completes one revolution, a certain number of pulses are generated through the A and B phase outputs of the Hall effect sensors. By counting the number of pulses within a certain time period, we can calculate the instantaneous speed of the motor.

There are three types of pulse counting modes:

- 1) The counter counts on the rising or falling edge of the pulses outputted by T1.
- 2) The counter counts on the rising or falling edge of the pulses outputted by T2.

- 3) The counter counts on both the rising and falling edges of the pulses outputted by both T1 and T2.

The third mode is commonly used, it enhances the accuracy of measurement. By determining whether the counter is incrementing or decrementing, we can determine the direction of rotation of the motor.

3.2 Coefficient

◆ ARR

when a pulse is encountered, the counter increments by one. When the count reaches the reload value, an overflow interrupt is generated. The counter is then cleared and starts counting again from zero. Therefore, the reload value represents the maximum value for pulse counting by the counter.

◆ PSC

Originally, the timer was used for timing or PWM output, and the prescaler factor was based on the 32-bit chip's internal clock. The formula used was $CK_CNT = TIMxCLK / (PSC + 1)$, where PSC represents the prescaler factor. However, when the timer is used for encoder speed measurement, the clock source is external.

3.3 Convert Pulse into Speed

When the motor completes a revolution, the encoder will receive $360 \times 4 / PSC$ pulses. Then use timer set m seconds and record the

When the motor completes one revolution, the encoder captures a total of $360 \times 4 / PSC$ pulses, where PSC represents the prescaler factor. Then, using a timer, we measure the duration of m milliseconds and record the number of timer overflows as n. The speed can be calculated using the formula $v = (n * ARR + \text{current count value}) / (360 * 4 / PSC) / m$, where ARR represents the auto-reload value of the timer.